

KT P0150 - SQL Aggregates AND OPERATORS - EXERCISE 2

1. Query: SELECT DISTINCT (department) FROM students;

Output:

department
IT
HR
FINANCE

2. Query: SELECT department, AVG(age) AS avg_age FROM students;
GROUP BY department;

Output:

department	avg_age
IT	20,5
HR	22
Finance	23

3. Query: SELECT department, COUNT(student_id) AS student_count
FROM students

GROUP BY department

HAVING student_count > 1

Output:

department	student_count
IT	2
HR	2

4. Query: SELECT student_id, name, age, department FROM Students
WHERE age BETWEEN 21 AND 23;

Output:

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. Query: SELECT student_id, name, age, department FROM Students
WHERE department IN ('IT', 'HR')
AND age > 21;

Output :

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6. Query: SELECT department, SUM(credits) AS total_credits FROM courses
GROUP BY department
HAVING total_credits > 5;

Output :

department	total_credit
IT	11

7. Query: SELECT course_id, course_name, department, credits FROM courses
WHERE ~~credits < 4~~ credits != 4;

Output :

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. ~~SELECT course_id, course_name, credits FROM courses
ORDER BY~~

8. Query: SELECT course_id, course_name, credits FROM courses
ORDER BY credits DESC;

Output :

course_id	course_name	credits
102	Python IT	4
103	Data Science IT	4
101	SQL basics	3
105	Statistics	3
104	Excel	2

9. Query: SELECT MAX(grade) AS max-grade, MIN(grade) AS min-grade,
AVG(grade) AS avg-grade FROM enrollments;

Output:	max-grade	min-grade	avg-grade
	90	78	84,6

10. Query: SELECT course-id, COUNT(enrollment-id) AS enrollment-count
FROM enrollments;

Output:	course-id	enrollment-count
	101	1
	102	1
	103	1
	104	1
	105	1

11. Query: SELECT department, SUM(salary) AS total-salary, SUM(bonus) AS total-bonus
FROM salaries
GROUP BY department;

Output:	department	total-salary	total-bonus
	IT	122 000	10 500
	HR	109 000	7 500
	Finance	70 000	6 000

12. Query: SELECT departments, AVG(salary) AS avg-salary FROM salaries
WHERE avg-salary > 55 000;

Output:	department	avg-salary
	IT	61 000

13. Query: SELECT employee-id, name, salary, bonus, (salary + bonus) AS total-compensation
FROM salaries

WHERE total-compensation > 60 000;

Output:	employee-id	name	salary	bonus	total-compensation
	1	Tom	60 000	5 000	65 000
	3	Spike	70 000	6 000	76 000
	4	Tyke	62 000	5 500	67 500

14. ~~SELECT~~ Query: SELECT department, SUM(budget) AS total-budget,
AVG(budget) AS avg-budget FROM projects

~~Output:~~ GROUP BY department
HAVING avg-budget > 70 000 ;

Output:

department	total-budget	avg-budget
IT	270 000	135 000
Finance	80 000	80 000

15. Query: SELECT project-id, project-name, department, budget FROM projects
WHERE department != 'Marketing'

AND budget BETWEEN 50 000 AND 120 000 ;

Output:

project-id	project-name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000